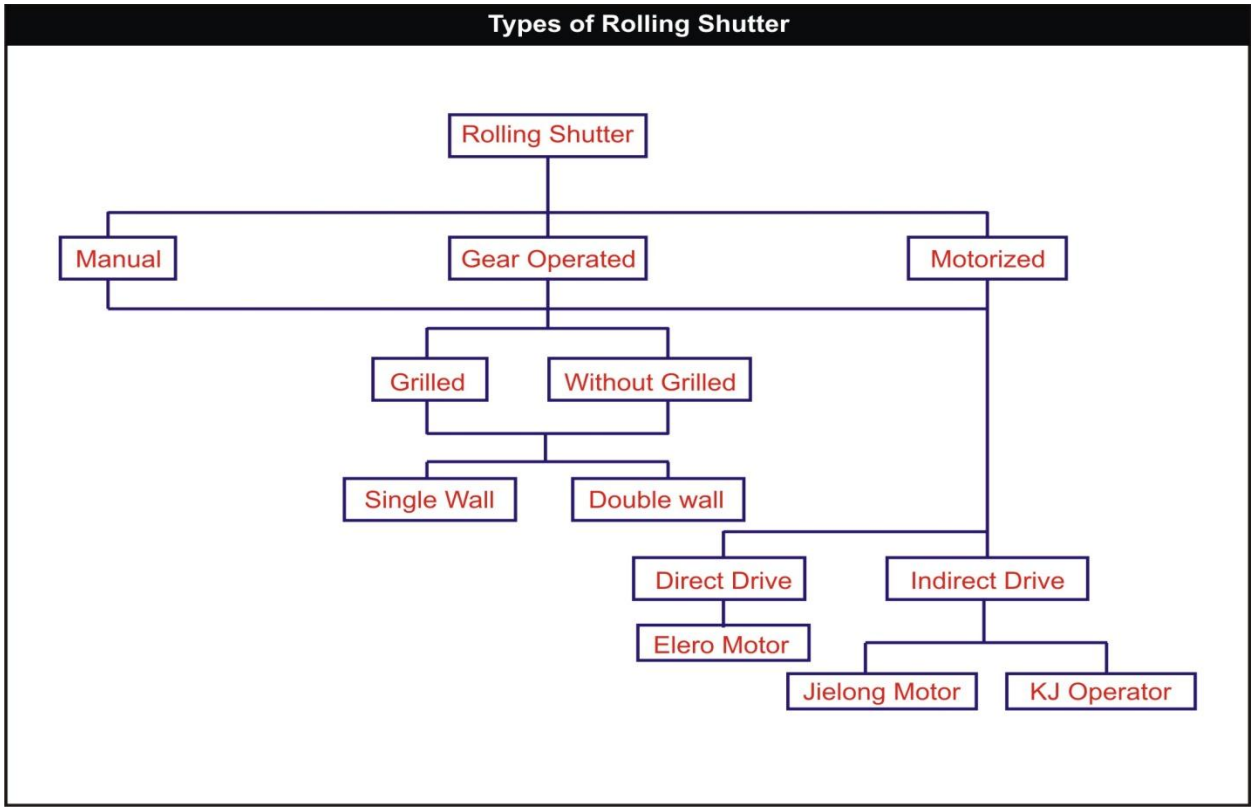
NIHVA
NIHVA Technologies Pvt. Ltd.

What is rolling shutter?

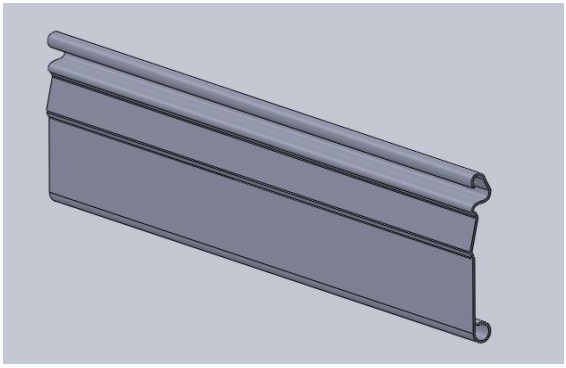
A roller shutter is a type of door or window shutter consisting of many horizontal slats (or sometimes bars or web systems) hinged together. The door is raised to open it and lowered to close it. On large doors, the action may be motorized. It provides protection against wind and rain.



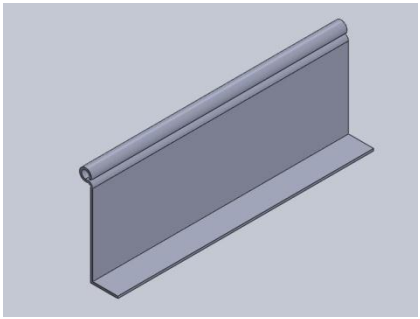
Design data sheet for Motorized rolling shutter

Important part of rolling shutter:-

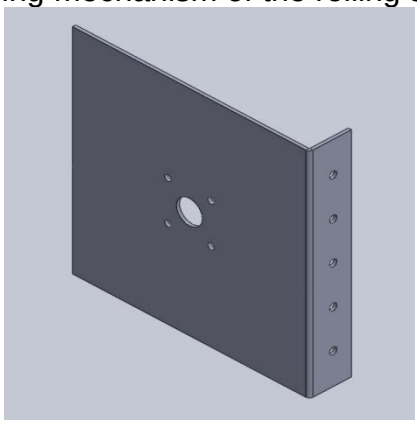
1. Lath or slat -The individual rolled interlocking laths or slats with which the rolling shutter curtain is assembled.



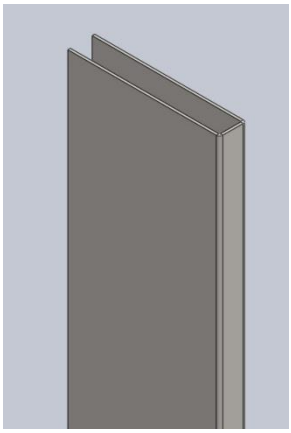
2.Bottom Lock Plate - The fabricated bar inserted at the bottom of rolling shutter curtain, so as to lie against the sill, including the slide bolts, Pulling handles, etc.



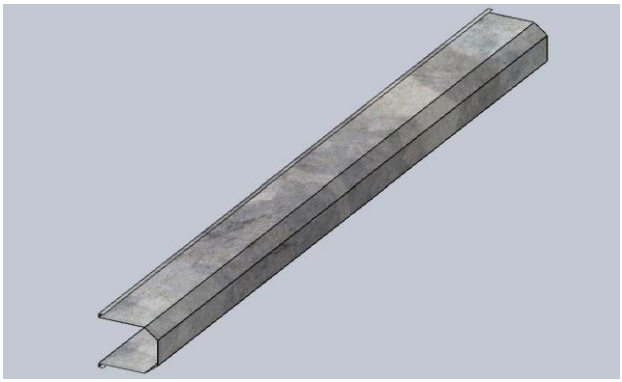
3. Bracket Plates -The supporting plates at either end on the top, together with the U-shaped clamps supporting the entire moving mechanism of the rolling shutter.



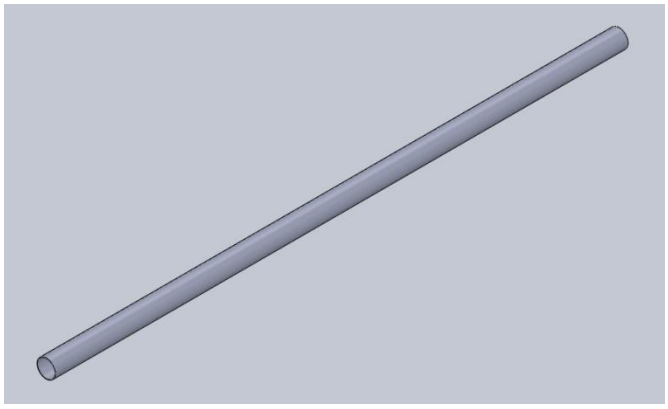
4. Guide Channels -- The channels on either side in which the shutter moves up and down.



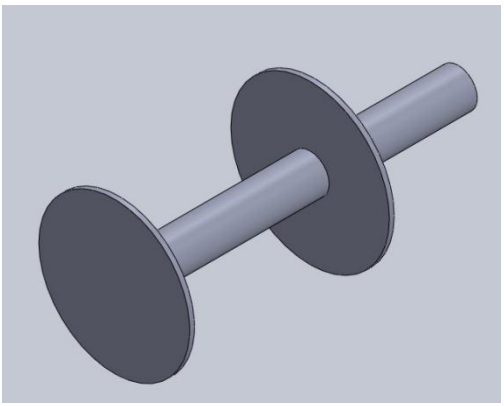
5. Hood Cover - A sheet metal cover bent into a suitable shape for covering the roller.

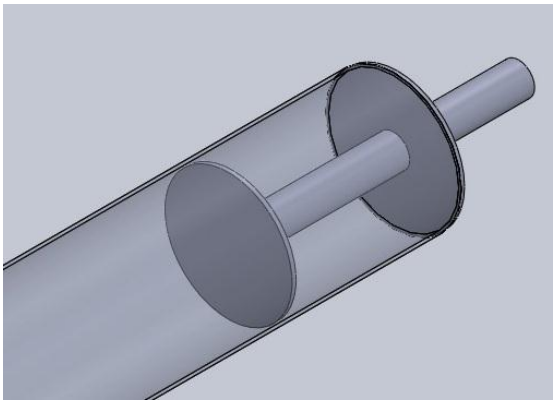


6. A) Pipe shaft – It is main shaft on which curtain rolls.

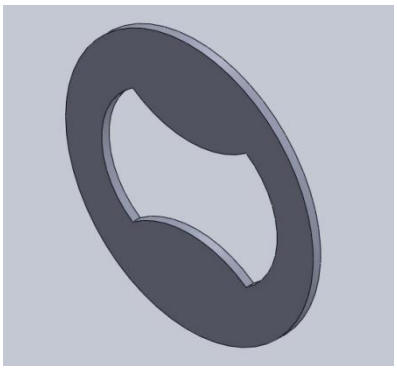


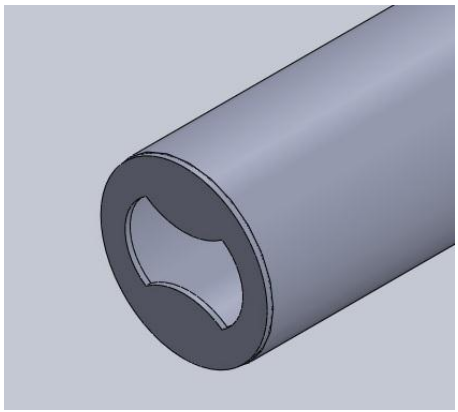
b) Flange shaft- It is shaft having two circular plates out which one will remain in pipe Shaft for safety in case of failure and other will be welded on pipe shaft.





c) Motor flange- It is flange has to be welded on other side of pipe shaft. Motor drive is attached To the pipe shaft through motor flange.

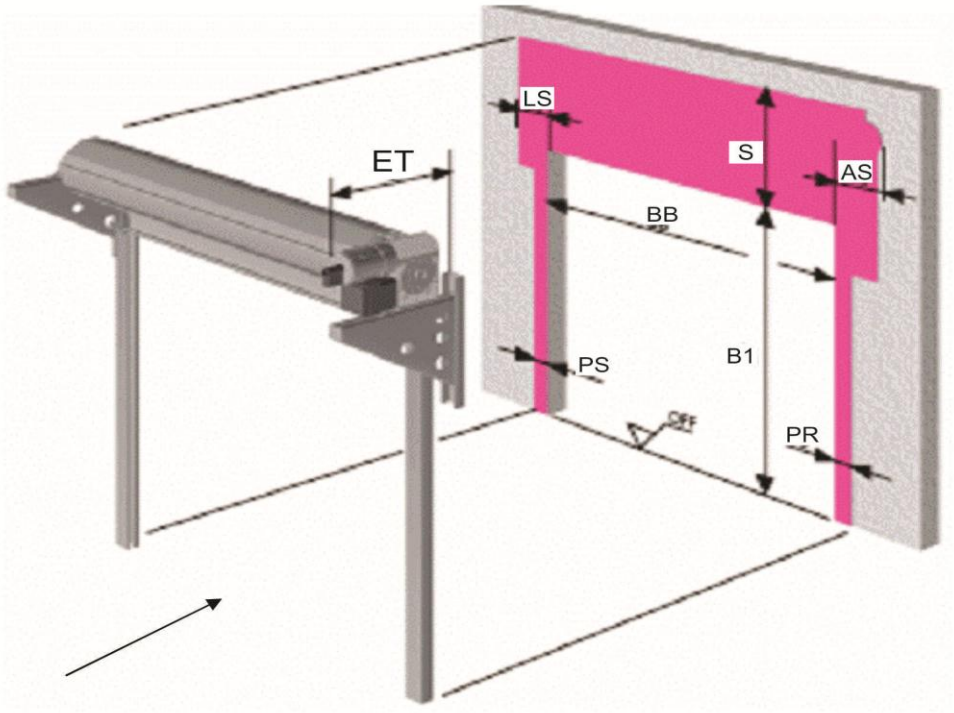




Check list for rolling shutter site

NIHVA
NIHVA Technologies Pvt. Ltd.

ROLLING SHUTTER MEASUREMENT SHEET



Inside room

Sr	Unit	Dimensions in mm
1	LS (left)	300
2	AS (right)	600
3	S (vertical up)	500
4	BB	Clear opening area
5	B1	Clear opening area
6	PS (left)	150
7	PR (right)	500
8	ET	750

Notes:

All 3 sides should be flush with each other
Wall or structural members required horizontally / vertically to support guide channels / shaft of the rolling shutter

+91 2066 3040 33 +91 85 51091 333 sales@nihva.com www.nihva.com

Design of rolling shutter

1. Selection for guide channel – The depth and width of the guide channel shall be as under:-

Clear width of shutter	Depth of guide as per IS6248-1979	Depth of guide as per NIHVA recommendation
Up to 3.5m	65mm	75mm
3.5m to 8m	75mm	100mm to 150mm
8m and above	100mm	150mm

2. Bracket plate - The bracket plate shall be fabricated out of mild steel of 3.15mm thickness (minimum), thicker plates may be used depending upon the height of shutter.

Clear Height in m	Size of bracket plate in mm
Up to 2.3m	300x300x3.15
2.3 to 2.6	325x325x3.15
2.3 to 3.0	350x350x3.15
3.0 to 3.5	375x375x3.15
3.5 to 4.5	400x400x6
4.5 to 5.5	450x450x6
5.5 to 6.5	500x500x6
6.5 to 8	550x550x8

3. **Selection for motor (indirect type Jielong make)**

For motor selection we should know how much weight of the shutter. The tentative weight for shutter considered as 25kg/sq.m.

Weight of shutter = Width (in m) x height (in m) x 25kg/sq.m

For example, if W= 5m & H=5m

Then approximate weight of shutter will be = 5x5x25

= 625 kg

Then select motor from following table i.e. Jielong ECR 800

JL–ECR产品系列主要型号、规格及技术参数表
JL-ECR phase motor specifications and technical parameters

型号规格 Model	额定输出转矩 (N·m) Rated output torque(N·m)	额定输出转速 (r/min) Rated output speed(r/min)	电机功率 (W) Motor power (W)	额定电压/频率 (V/Hz) Rated voltage/ frequency (V/Hz)	额定电流 (A) Rated current (A)	持续工作时间 (min) Con inued work time (min)	标定卷升参数 Standard roll up parameter			
							卷筒外径 (mm) Drum dube outer diameter (mm)	最大提升力 (kg·f) Maximum Lifting power (kgf)	最大提升高度 (m) Maximum lifting height (m) 帘片厚度 (mm) Profile thickness(mm)	
									15	20
单相交流电动卷帘开门机 ECR Single phase motor specifications and technical parameters										
ECR—112—1P	112	5.0	70	220/50	2.8	8	114	200	4	4.5
ECR—168—1P	168	5.0	250	220/50	2.8	8	114	300	6	7
ECR—225—1P	225	4.5	180	220/50	1.8	8	114	400	6	7
ECR—343—1P	343	4.8	370	220/50	3.2	8	140	500	6	7
ECR—412—1P	412	4.8	370	220/50	3.3	8	140	600	6	7
ECR—647—1P	647	4.0	400	220/50	4.7	8	165	800	7	7
ECR—809—1P	809	3.5	400	220/50	4.9	8	165	1000	6	7
ECR—1617—1P	1617	3.3	700	220/50	8.2	8	219	1500	9	10
三相交流电动卷帘开门机 JL-ECR three phase motor specifications and technical parameters										
ECR—168—3P	168	5.0	250	380/50	1.5	13	114	300	6	7
ECR—225—3P	225	4.2	180	380/50	0.75	13	114	400	6	7
ECR—343—3P	343	4.8	250	380/50	1.5	13	140	500	6	7
ECR—412—3P	412	4.8	250	380/50	1.5	13	140	600	6	7
ECR—647—3P	647	4.0	370	380/50	2.3	13	165	800	7	8
ECR—809—3P	809	3.5	370	380/50	2.8	13	165	1000	6	
ECR—1401—3P	1401	4.4	850	380/50	3.2	13	219	1300	7	8
ECR—1617—3P	1617	5.2	1000	380/50	3.8	13	219	1500	8	9
ECR—2450—3P	2450	3.6	1000	380/50	4.1	13	245	2000	12	13

注：若客户对卷帘门机的“最大提升力”、“最大提升高度”、“卷筒外径”及“输出转速”等使用性能参数及其参数组合，有超出主要型号规格配置之外的特殊变化的需求，可向本公司定制特配(拓展型)产品。
Remark: If you need any special design we can make it for you as your request.

4. Selection of pipe shaft-

We can simply select pipe shaft from following table relevant to motor selection.

Nominal diameter of Shaft/MS pipe	Outer diameter of Pipe in mm	Maximum lifting Capacity in kg	Nihva recommended motor(Jielong make)
(DN 100mm) 4”	114	200-400	ECR 400
(DN 125mm) 5”	140	500-600	ECR 600
(DN 150mm) 6”	165	800-1000	ECR 800/ ECR 1000
(DN 200mm) 8”	219	1000 Up to 1500	ECR1000/ECR1500
(DN 225mm) 9”	245	1500-2000	ECR2000

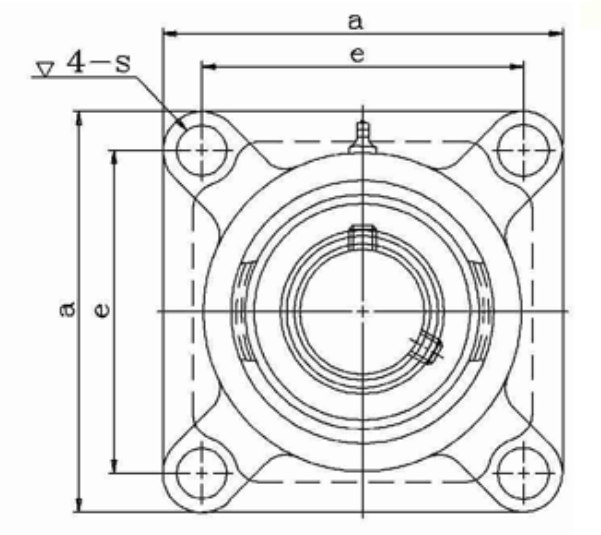
Let's design a shutter of size 5m (Width) x 5m (Height) - case study for SMCC, Gujarat. We have observed sag in shutter so let's keep guide height 5300mm instead of 5000 to Avoid sag in shutter.

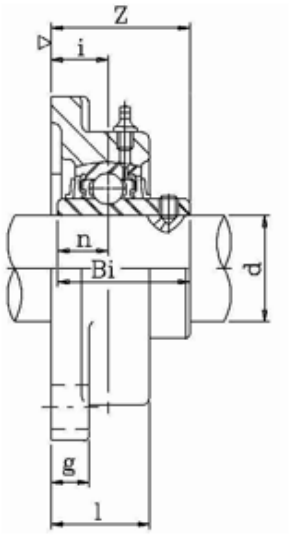
1. Guide channel- From guide selection table we can select 100mm guide also but as per customer demand or safety purpose select 150mm guide width.
Hence, w=150mm & L=5300mm
2. Slat- Total width of shutter= Clear width + Width of both shutter
= 5000mm+ 300mm
=5300mm -100mm clearance for sliding of slat
=5200mm.
3. Bracket plate- As height of shutter is actually 5300mm then select bracket dimensions from table which is 450x450x6mm.

4. Selection of motor- first calculate the approximate weight of rolling shutter.
Weight= 5.3mx5mx25 sq.m
= 662.5kg
Then select motor from table i.e. Jielong model ECR800 capacity- 800kg

5. Bearing- In motorized rolling shutter only one bearing required at non motor side. Generally 40mm solid bar is used for flanged shaft with two circular plate out which one will be in pipe shaft other will be welded outer surface of pipe shaft.
For 40mm shaft select - NTN CM UCF208D1 or same spec. of other make.
As we have selected DN (150mm) 6" pipe of outer diameter 168mm and inner diameter 161.5mm. So diameter of circular plates of 5mm THK. Should be 166mm & 158mm respectively.

6. Hood cover- It is the cover which is mounted on ms brackets to protect shutter and shaft assembly from water & dust. Another application of hood cover is for better Aesthetic of shutter.







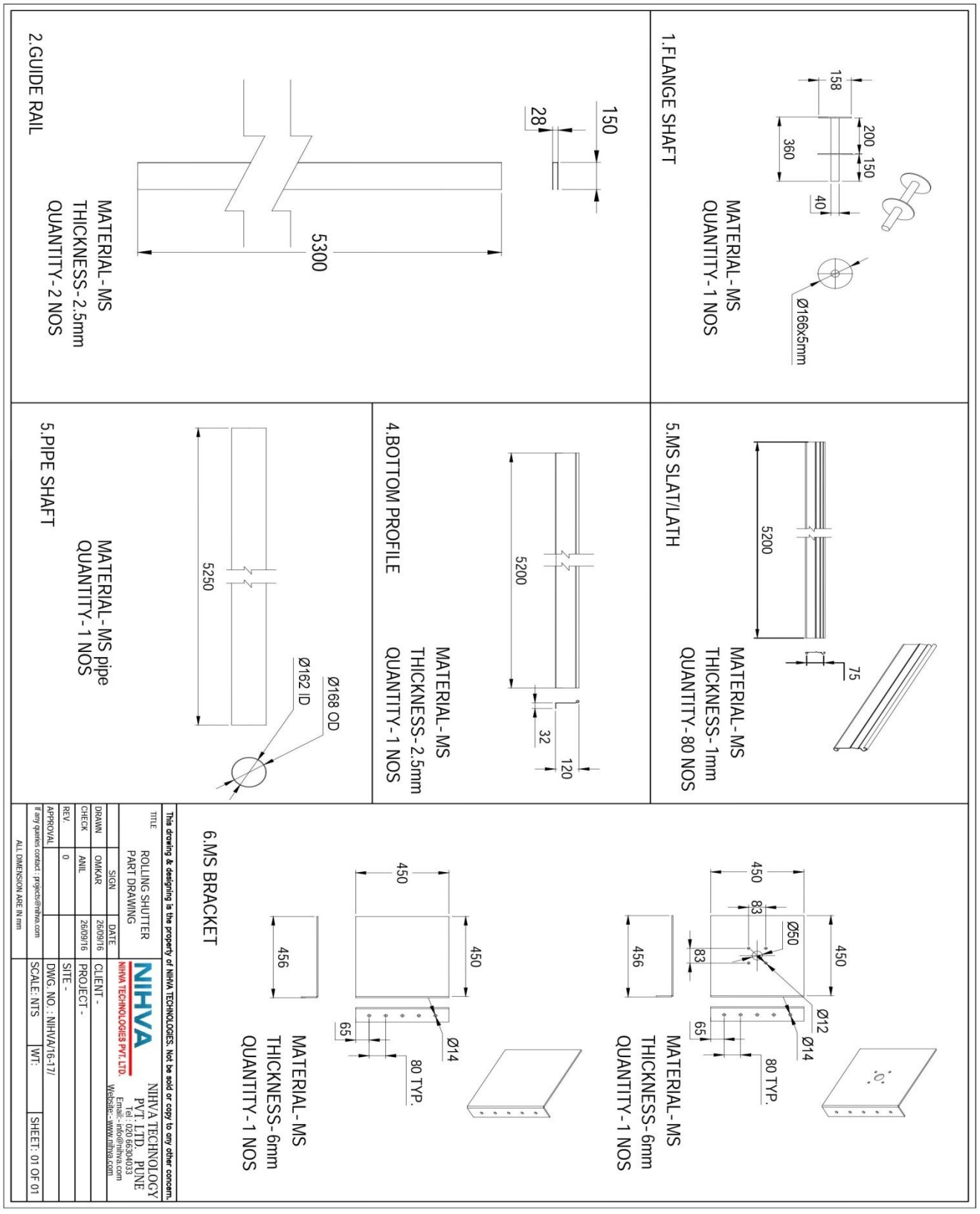
SR type Seal

UCF 208-40MM Dimensions

Unit	Shaft Dia	a	e	i	g	l	s	z	Bi	n	Bolt Size
inch		5 1/8	4 1/64	53/64	5/8	1 13/32	5/8	2 1/32	1.937	0.7480	1/2
mm	40	130	102	21	16	36	16	51.2	49.2	19	M14

For example 1000 kg motor. We need 40 mm bearing ID.

Following drawings are attached for manufacturing of s



This drawing & designing is the property of NIHVA TECHNOLOGIES. Not be sold or copy to any other concern.

TITLE: HOOD COVER

NIHVA TECHNOLOGIES PVT. LTD.

NIHVA TECHNOLOGY PVT. LTD., PUNE
Tel : 020 66304033
Email:- info@nihva.com
Website:- www.nihva.com

	SIGN	DATE
DRAWN	OMKAR	26/09/16
CHECK	ANIL	26/09/16
REV:		
APPROVAL		

If any queries contact : project@nihva.com

CLIENT -

PROJECT -

SITE -

DWG. NO. : NIHVA/16-17/

ALL DIMENSION ARE IN mm

SCALE: NTS | WT: | SHEET: 01 OF 01

BILL OF MATERIAL FOR MOTORIZED ROLLING SHUTTER SIZE- 5x5m

SR NO	PART NAME	MATERIAL	SPECIFICATION	QUANTITY	UNIT COST	BASIC COST	COST WITH TAX& TRANS.
1	SLAT	MS	5200mm	80	75/KG	27000	
2	GUIDE	MS	5300mm	2	60/KG	4200	
3	BRACKETS	MS	450x450x6mm	2	1000/PC	2000	
4	BEARINGS	CAST IRON	NTN CM UCF208D1	1	1200/PC	1200	1200
5	FLAT	MS	30x3000x5mm	1			
6	SHAFT FLANGE	MS	40mm BRIGHT BAR	1		700	
			WITH FLANGE				
7	SHAFT PIPE	MS	6" INCH OD-168mm	1		3700	
			THICKNESS-3.4mm				
			LENGTH-5250mm				
8	MOTOR	JIELONG MAKE	1000 KG	1		13100	
9	ANCHOR F.	MS	12X100mm	20		480	
10	KLITS & RIVETS						
11	NUT& BOLTS	MS	M10X60	10		70	
12							
13	GASKET	RUBBER	LENGTH-5000mm	1	140/m	700	
14	MOTOR CHAIN	STEEL	8000mm	1		680	
15	COATING	PAINT	RED OXIDE	15 LIT.	150/LIT	2250	
16	ANGLE	MS	35x35mm	1			
17	HOOD COVER	GI	5600mm	1		4000	
TOTAL COST							